

MySQL Assignment 2

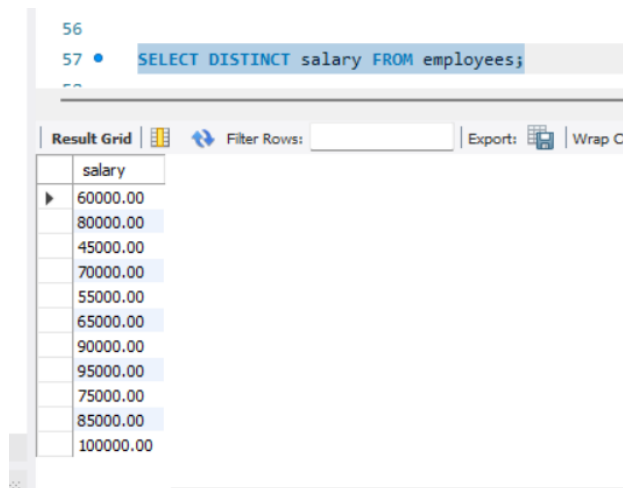
Querying Data

Tasks

Clause & Operators:

1. DISTINCT VALUES:

A query to retrieve distinct salaries from the Employees table.



The screenshot shows a MySQL query editor with the following SQL query:

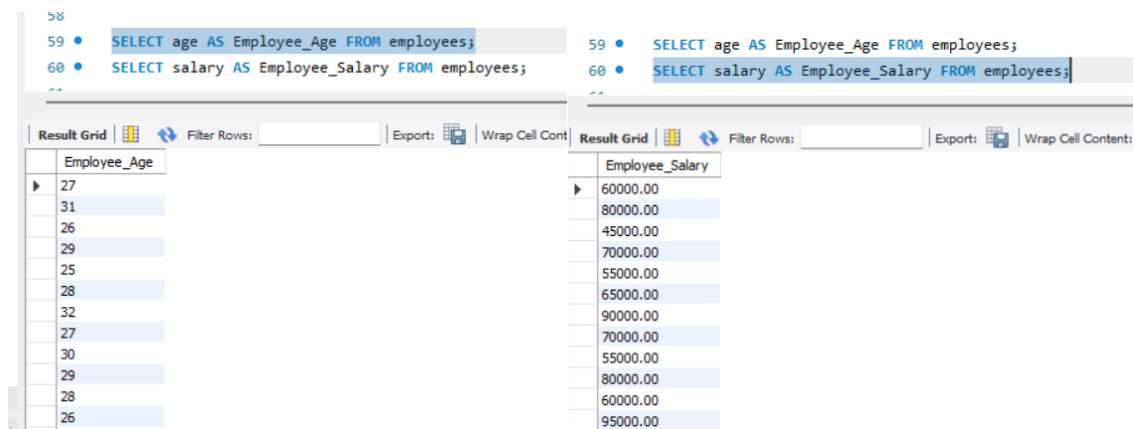
```
56  
57 • SELECT DISTINCT salary FROM employees;  
58
```

Below the query editor is the 'Result Grid' showing the results of the query. The grid has a single column labeled 'salary' and contains 15 rows of distinct salary values.

salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

2. ALIAS (AS):

Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.



The screenshot shows two side-by-side MySQL query editors and their respective result grids.

Left Editor:

```
58  
59 • SELECT age AS Employee_Age FROM employees;  
60 • SELECT salary AS Employee_Salary FROM employees;  
61
```

Left Result Grid:

Employee_Age
27
31
26
29
25
28
32
27
30
29
28
26

Right Editor:

```
59 • SELECT age AS Employee_Age FROM employees;  
60 • SELECT salary AS Employee_Salary FROM employees;  
61
```

Right Result Grid:

Employee_Salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
70000.00
55000.00
80000.00
60000.00
95000.00

3. WHERE CLAUSE & OPERATORS:

* Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
62
63 • SELECT * FROM employees
64 WHERE salary > 50000 AND hire_date <= '2016-01-01' ;
65
```

Result Grid

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

* Find the employee whose designation is missing and fill it with "Data Scientist".

```
68
69 • UPDATE employees
70 SET designation = 'Data Scientist'
71 WHERE designation IS NULL ;
72
73 • SELECT * FROM employees ;
74
```

Result Grid

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5017	Kabir Nair	M	28	2017-09-11	IT Support Specialist	12	2	80000.00
5018	Ishan Mishra	M	25	2017-11-20	Research Scientist	9	3	75000.00
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	7	3	85000.00
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00
5024	Aryan Shetty	M	27	2018-11-05	Product Manager	5	1	95000.00
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	3	4	100000.00
5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	3	2	75000.00
5027	Ananya Paul	F	32	2019-05-20	Content Writer	6	3	60000.00
5028	Anaya Kapoor	F	26	2019-07-05	Event Coordinator	6	1	60000.00

employees 9 x

[illegible]

3. AGGREGATE FUNCTIONS:

- Calculate the sum of all salaries in the Finance department.

```
3 • SELECT sum(salary) FROM employees
4 WHERE designation LIKE '%Finance%';
5
```

sum(salary)
85000.00

- Find the minimum age among all employees.

```
86 • SELECT MIN(age) FROM employees ;
87
```

MIN(age)
25

4. GROUP BY:

- List the maximum salary for each location.

```
88 • SELECT location_id,MAX(salary) AS MAX_salary FROM employees
89 GROUP BY location_id ;
90
```

location_id	MAX_salary
1	95000.00
2	95000.00
3	90000.00
4	100000.00

- Calculate the average salary for each designation containing the word 'Analyst'.

```

90
91 • SELECT designation, AVG(salary) AS AVG_salary FROM employees
92 WHERE designation LIKE '%Analyst'
93 GROUP BY designation ;
94
95

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 		
	designation	AVG_salary
▶	Data Analyst	60000.000000
	Marketing Analyst	60000.000000
	Supply Chain Analyst	60000.000000
	Financial Analyst	85000.000000
	Finance Analyst	85000.000000
	Business Analyst	75000.000000
	Quality Assurance Analyst	80000.000000

5. HAVING:

- Find departments with less than 3 employees.

```

96 • SELECT department_id, COUNT(*) AS employee_count FROM employees
97 GROUP BY department_id
98 HAVING employee_count < 3
99 ORDER BY employee_count DESC;
100
101

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 		
	department_id	employee_count
▶	5	2
	6	2
	7	2
	11	2
	1	1
	9	1
	13	1

- Find locations with female employees whose average age is below 30.

```

102 • SELECT location_id,ROUND(AVG(age)) AS Location_AVG_below_30 FROM employees
103 WHERE gender='F'
104 GROUP BY location_id
105 HAVING AVG(age)<30;
106

```

location_id	Location_AVG_below_30
1	27
2	27
4	29

Joins:

1. INNER JOIN:

- List employee names, their designations, and department names where employees are assigned to a department.

```

107
108 • SELECT departments.department_name, employees.employee_name, employees.designation
109 FROM departments
110 INNER JOIN employees
111 ON departments.department_id = employees.department_id;
112

```

department_name	employee_name	designation
Business Development	Anvi Chaudhary	Business Development Executive
Business Development	Kiaan Desai	Sales Executive
Content Creation	Ananya Paul	Content Writer
Content Creation	Anaya Kapoor	Event Coordinator
Customer Support	Aaradhya Iyer	Customer Support Executive
Customer Support	Arnav Rao	Customer Success Manager
Customer Support	Vihaan Mohan	Supply Chain Analyst
Data Science	Vihaan Singh	Data Analyst
Data Science	Avni Iyengar	Data Scientist
Data Science	Vivaan Singh	Business Analyst
Design	Kiara Malhotra	Data Scientist
Design	Dhruv Shetty	UI Developer

Result 20 x

2. LEFT JOIN:

- List all departments along with the total number of employees in each department, including departments with no employees.

```
113 • SELECT departments.department_name,  
114 COUNT(employees.employee_id) AS total_employees  
115 FROM departments  
116 LEFT JOIN employees  
117 ON departments.department_id = employees.department_id  
118 GROUP BY departments.department_name;
```

Result Grid	
Filter Rows:	Export: Wrap Cell Content:
department_name	total_employees
Business Development	2
Content Creation	2
Customer Support	3
Data Science	3
Design	3
Finance	2
Human Resources	3
IT	4
Marketing	3
Operations	1
Product Management	2
Research and Develo...	1

3. RIGHT JOIN:

- Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```
120 • SELECT location.location,  
121 employees.employee_name  
122 FROM location  
123 RIGHT JOIN employees  
124 ON location.location_id = employees.location_id;
```

Result Grid	
Filter Rows:	Export: Wrap Cell Content:
location	employee_name
Pune	Vihaan Singh
Chennai	Reyansh Singh
Bangalore	Aaradhya Iyer
Hyderabad	Kiara Malhotra
Chennai	Anvi Chaudhary
Bangalore	Dhruv Shetty
Hyderabad	Anushka Singh
Pune	Diya Jha
Hyderabad	Kiaan Desai
Pune	Atharv Yadav
Chennai	Saanvi Patel
Bangalore	Myra Verma

4. CROSS JOIN:

- Show all possible combinations of departments and locations.

```
125
126 • SELECT departments.department_name,
127     location.location
128 FROM departments
129 CROSS JOIN location;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

	department_name	location
▶	Business Development	Pune
	Business Development	Hyderabad
	Business Development	Bangalore
	Business Development	Chennai
	Content Creation	Pune
	Content Creation	Hyderabad
	Content Creation	Bangalore
	Content Creation	Chennai
	Customer Support	Pune
	Customer Support	Hyderabad
	Customer Support	Bangalore
	Customer Support	Chennai

5. SELF JOIN:

- Show pairs of employees working in the same department, excluding self-pairs.

```
131 • SELECT e1.employee_name AS employee_1,
132     e2.employee_name AS employee_2
133 FROM employees e1
134 JOIN employees e2
135 ON e1.department_id = e2.department_id
136 AND e1.employee_id < e2.employee_id;
```

Result Grid | Filter Rows: | Export: | Wrap

	employee_1	employee_2
▶	Vihaan Singh	Avni Iyengar
	Vihaan Singh	Vivaan Singh
	Reyansh Singh	Atharv Yadav
	Reyansh Singh	Kabir Nair
	Reyansh Singh	Arjun Kumar
	Aaradhya Iyer	Arnav Rao
	Aaradhya Iyer	Vihaan Mohan
	Kiara Malhotra	Dhruv Shetty
	Kiara Malhotra	Diya Jha
	Anvi Chaudhary	Kiaan Desai
	Dhruv Shetty	Diya Jha
	Anushka Singh	Saanvi Patel

Windows function:

- Write a window function query to rank employees by salary using rank().

```
38 • SELECT employee_name,  
39 salary,  
40 RANK() OVER(PARTITION BY department_id  
41 ORDER BY salary DESC) AS salary_rank  
42 FROM employees;  
43
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

employee_name	salary	salary_rank
Aarav Nair	90000.00	1
Anushka Singh	90000.00	1
Anika Paul	70000.00	2
Saanvi Patel	60000.00	3
Avni Iyengar	100000.00	1
Vivaan Singh	75000.00	2
Vihaan Singh	60000.00	3
Zoya Khan	100000.00	1
Aadhya Iyengar	60000.00	2
Ishika Patel	55000.00	3
Aryan Shetty	95000.00	1
Sara Iyer	90000.00	2

- Write a window function query to rank employees by salary within each department using DENSE_RANK()

```
143  
144 • SELECT employee_name,  
145 department_id,  
146 salary,  
147 DENSE_RANK() OVER(PARTITION BY department_id  
148 ORDER BY salary DESC) AS salary_rank  
149 FROM employees;  
150
```

Result Grid | Filter Rows: | Export: | Wrap Cell Center

	employee_name	department_id	salary	salary_rank
▶	Aarav Nair	1	90000.00	1
	Anushka Singh	2	90000.00	1
	Anika Paul	2	70000.00	2
	Saanvi Patel	2	60000.00	3
	Avni Iyengar	3	100000.00	1
	Vivaan Singh	3	75000.00	2
	Vihaan Singh	3	60000.00	3
	Zoya Khan	4	100000.00	1
	Aadhya Iyengar	4	60000.00	2
	Ishika Patel	4	55000.00	3
	Aryan Shetty	5	95000.00	1
	Sara Iyer	5	90000.00	2

- Write a window function query, Running total salary by department.

```
51 • SELECT employee_name,  
52     department_id,  
53     salary,  
54     SUM(salary) OVER(PARTITION BY department_id  
55     ORDER BY employee_id) AS running_total_salary  
56     FROM employees;  
57
```

Result Grid			
Filter Rows:		Export:	Wrap Cell Content
employee_name	department_id	salary	running_total_salary
Aarav Nair	1	90000.00	90000.00
Anushka Singh	2	90000.00	90000.00
Saanvi Patel	2	60000.00	150000.00
Anika Paul	2	70000.00	220000.00
Vihaan Singh	3	60000.00	60000.00
Avni Iyengar	3	100000.00	160000.00
Vivaan Singh	3	75000.00	235000.00
Zoya Khan	4	100000.00	100000.00
Ishika Patel	4	55000.00	155000.00
Aadhya Iyengar	4	60000.00	215000.00
Aryan Shetty	5	95000.00	95000.00
Sara Iyer	5	90000.00	185000.00

Result 31 ×